

João Victor Mendes

Production Support Engineer (PSE) / SRE | Cloud · Observability · Database Reliability

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União da Vitória, PR, Brazil (UTC-3)

PROFESSIONAL SUMMARY

Silver Medalist, Brazilian Mathematics Olympiad (OBMEP, Top 0.01% of 18M+ participants nationally). Translates extreme analytical rigor to structural incident response, observability, and reliability engineering in high-scale AI and FinTech platforms. Specialized in deep distributed systems diagnostics, mitigating database saturation bottlenecks (AWS Aurora MySQL, Kubernetes, Airflow), and structural telemetry noise elimination, with a track record of cutting metrics ingestion by over 70% and leading critical production post-mortems.

TECHNICAL SKILLS

- **Cloud & Orchestration:** AWS (Aurora MySQL/RDS, EKS, ECS, SQS, SSM, CloudWatch, Secrets Manager, SES), Kubernetes (Karpenter, ArgoCD, KEDA, Topology Spread Constraints), Docker.
- **Observability & SRE:** New Relic (NRQL, NerdGraph API, baseline/alerting), OpenTelemetry (OTel Collector), P1/P2 Incident Response, ITIL-aligned RCA / Post-Mortems, Detection Engineering & Runbooks, SLA.
- **Databases & Storage:** AWS Aurora MySQL (Performance Insights, connection pooling, parameter groups), PostgreSQL (WAL, replication slots, bloat, EXPLAIN ANALYZE, pg_class), AWS Redshift, Redis / Valkey.
- **IaC, Automation & Languages:** Terraform, Terragrunt, Atlantis, Python, Bash, PowerShell, C#/.NET, SQL, Git.
- **Languages:** English (Certified B2 Full Professional Proficiency, BRASAS), Portuguese (Native).

PROFESSIONAL EXPERIENCE

Monest | Credit Recovery FinTech & AI Platform Aug 2026 – Present
Production Support Engineer (PSE) / SRE Remote, Brazil

Engineering reliability, observability, and cloud infrastructure for a high-scale distributed platform powering AI agents (MIA), VoIP dialers (Jambonz), and cloud microservices.

- **Database Saturation Diagnostics & Pool Capacity (Aurora MySQL):** Led investigation into critical production database saturation (writer at 98.8% CPU, 1,000 connection ceiling hit dropping 383 jobs across 27 companies in AI agent queues); quantified connection-pool fan-out across microservices (23 instances × 50 = 1,150 potential connections vs. 1,000 hard cap) and unbatched queries, shifting engineering priorities toward RDS Proxy adoption and parameter-group decoupling.
- **Zero Noise Observability & Structural Alerting Flaws (New Relic / NerdGraph):** Audited and classified 467 production alert conditions via NerdGraph API; identified an unsegmented policy generating ~4,900 incidents/month (~40% of account volume) and discovered a structural mathematical flaw in baseline alerting (ALL duration > aggregation window on sparse crons) that silently suppressed notifications, formally incorporated into the company-wide observability RFC.
- **Data Pipeline Outage Response & Detection Engineering (Airflow / Docker / SSM):** Diagnosed and led post-mortem for a 3h45 data pipeline outage caused by Docker ENOSPC during a 108k voice-records batch; bypassed SSH blockers using AWS Systems Manager (SSM) without admin rights, formulated a scheduler deadman alert reducing MTTD from 2h02 to ~10 minutes, and resolved a multi-mount disk alerting bug in IaC where averaging masked critical 99.89% root volume saturation.
- **Scheduler Outage Resolution & Host Persistence (Cronicle / Systemd / IaC):** Traced root cause of a 13-hour scheduler outage (154 missed runs across 4 operational pillars) undetected by 472 alert conditions because health checks monitored nginx rather than scheduler execution; engineered permanent host recovery via systemd service units, fail-safe fstab mounts (nofail), and pm2 boot persistence, validated in production during a measured 179-minute safe window without downtime.
- **K8s Resiliency & AI Stack SPOF Elimination (EKS / Karpenter / ArgoCD):** Resolved a 21-minute Langfuse outage with zero LLM observations recorded after Spot node interruption terminated single-replica web, worker, and stateful Valkey pods; restructured deployment via ArgoCD enforcing strict topologySpreadConstraints (whenUnsatisfiable: DoNotSchedule), preventing single-node bin-packing by Karpenter and ensuring cross-node high availability in production.
- **RDS Alerting Governance & Incident Detection (Terraform / New Relic):** Authored 8 production Aurora monitoring policies in Terraform with thresholds derived from 336 measured 30-minute buckets over 7 days; identified and eliminated a structural deduplication flaw that kept load alerts continuously open (breaching 37.8% of the time on writer and 53.6% on replica) and suppressed subsequent incident pages; the new alerts proactively detected critical connection storms 4 days after rollout.

Pedroso Automação | B2B fiscal compliance ERP SaaS
Technical Support Engineer | L2/L3 Infrastructure Recovery

June 2024 – Aug 2026

União da Vitória, PR

Direct L2/L3 support for 500+ corporate environments (1,500+ end-users). Maintained full independence on solo rotations, resolving P1/P2 incidents without escalation through deep structural analysis.

- **Eliminated the WAL retention risk class** across the fleet: built a Task Scheduler PowerShell monitor for PG12 that detects inactive orphaned replication slots and drops them when retention exceeds 2GB or inactivity exceeds 30 minutes, with Windows Event Log alerts. On PG13+ instances, complemented with `max_slot_wal_keep_size` as a hard server-side cap.
- **Automated 30-database cluster recovery** by developing a PowerShell script parallelizing state checks, clearing stale `postmaster.pid` files, and monitoring WAL recovery on boot; reduced recovery time from hours to <5 minutes.
- **Standardized PostgreSQL deployment on Windows** by scripting end-to-end installation, service configuration, and runtime hardening; cut manual setup time from ~2 hours to under 7 minutes across fleet environments.
- **Diagnosed 1GB/month disk bloat** through `pg_class` and `pg_toast`; traced root cause to encrypted file paths serialized as thousands of characters per audit log entry; reduced restore time 60% (17 min to 6:58) via `pg_restore -j 4`.
- **Engineered 70M-record ETL pipeline** cutting processing time 60% on a 15GB+ govt dataset; extended with client classification (MEI, Simples Nacional, Normal, invalid CNPJ), XML regime auditing (NF-e/NFC-e/NFS-e/CT-e), and PostgreSQL staging, surfacing the top 0.001% of active commercial leads.
- **Protected R\$500k in live inventory (~\$100k USD)** by reverse-engineering a legacy Firebird database with no documentation; replayed 20 years of transaction changelogs to reconstruct full inventory state with zero data loss, delivered the same day the schema was first seen.
- **Reduced fiscal diagnosis from 30+ min to <5 min** by developing a structured LLM-assisted triage workflow: direct extraction of tax rejection rules from SEFAZ XML with database context injection, eliminating developer escalations.
- **Established shared runbooks and mentoring culture**; produced 20+ video guides and ITIL post-mortems, reducing dependency on senior availability and accelerating onboarding.

Girafa Comunicação Interativa | Digital Agency
Cloud Infrastructure Analyst & Web Operations

Sep 2022 – Apr 2024

Remote, Brazil

Operated Cloud infrastructure and web operations for 70+ corporate clients (AWS Route 53, Registro.br, Locaweb).

- **Cloud Operations & Incident Resolution:** Managed corporate DNS, WAF, and zones across AWS Route 53 and CloudFront; served as international liaison with external registrars (GoDaddy, AWS, Namecheap) during complex migrations and isolated propagation/caching failures.
- **Security Incident Investigation:** Audited Git histories and access logs to identify security breaches; revoked compromised sessions and rotated credentials in high-priority security incidents.

EDUCATION & PEDIGREE

FGV — Data Science & Economics (2020–2022) Rio de Janeiro, Brazil
Full-ride OBMEP scholarship; EDA: 9.71. Left after two years to return home — the scholarship required full-time residence 1,000 km from family; chose hands-on engineering over completing the degree in Rio.

IFPR — Systems Analysis & Development (2023–2024) União da Vitória, Brazil
Grade A in Mathematics for Computing. Left to transition to full-time production engineering; found production incident work accelerated learning faster than the evening curriculum.

Silver Medalist — Brazilian Mathematics Olympiad (OBMEP, 2016) National
Top 0.01% of 18M+ participants nationally. Bronze Medalist (2014); 2x Honorable Mentions.